Extreme least singular values with redundancy, radial ensembles, and quantitative stability of phase retrieval

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June 28, 2026

Abstract

For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}$, let $G_m \in \mathbb{F}^{N_m \times m}$ be standard Gaussian, and assume $N_m/m \to \gamma > 1$. For

$$M^{\mathbb{F}}_{m,r}(G_m) = \min_{T \subset [N_m],\, |T|=m+r} \sigma_m((G_m)_T),$$

we prove that if $k_m = r_m + 1$ and $k_m \log m = o(m)$, then

$$\frac{k_m}{m} \log M^{\mathbb{F}}_{m,r_m}(G_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}, \qquad h(\gamma) = \gamma \log \gamma - (\gamma - 1)\log(\gamma - 1).$$

For fixed surplus r, this gives

$$\frac{1}{m} \log M^{\mathbb{F}}_{m,r}(G_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}.$$

The same exponential law holds for radial row ensembles with subexponential radii and for subexponentially anisotropic Gaussian rows. Applied to real phase retrieval with $N = 2m - 1 + r_m$, it yields

$$\frac{r_m + 1}{m} \log \omega(A_m) \xrightarrow{P} -\log 4, \qquad \omega(A_m) = 4^{-m/(r+1) + o_P(m)}$$

when $r_m \equiv r$ is fixed.

Keywords. Least singular value; Gaussian matrices; hard edge; phase retrieval; complement property; random frames; radial ensembles.

MSC 2020. 60B20, 15B52, 42C15, 94A12.

Contents

1 Introduction

The least singular value of a random matrix is a hard-edge observable: it measures how close the matrix is to losing rank. A single square Gaussian matrix has least singular value of polynomial size in the dimension, but the minimum over exponentially many dependent square submatrices of a taller matrix can be exponentially small. The square case was determined in [16]. If $G_m \in \mathbb{F}^{N_m \times m}$ has independent standard Gaussian entries and $N_m/m \to \gamma > 1$, then

$$\frac{1}{m} \log \min_{|T|=m} \sigma_m((G_m)_T) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}, \qquad h(\gamma) = \gamma \log \gamma - (\gamma - 1) \log(\gamma - 1).$$

In the real case, this implies the Gaussian base 1/4 in the Balan–Wang phase retrieval stability problem at the critical number $N = 2m - 1$ of measurements.

The purpose of this paper is to extend this result from square submatrices to row submatrices with a surplus number of rows. For an $N \times m$ matrix A over $\mathbb{F}$ and for $r \geq 0$, set

$$M^{\mathbb{F}}_{m,r}(A) = \min_{T \subset [N],\, |T| = m+r} \sigma_m(A_T).$$

The square case is $r = 0$. In that case the statement reduces to the square-submatrix law of [16]. The present paper does not use the square case as a black box. The new ingredients are the rectangular hard-edge exponent $d_{\mathbb{F}}(r+1)$, a $k = r+1$ marked-row construction, a two-normal estimate for k simultaneous marked constraints, and a good-overlap/bad-overlap second-moment decomposition that is uniform in the moderate range $k \log m = o(m)$. These additions are what produce the fixed- and moderate-redundancy phase-retrieval laws below.

The natural hard-edge heuristic is simple. A single $(m+r) \times m$ Gaussian matrix satisfies

$$\mathbb{P}\{\sigma_m \leq t\} = t^{d_{\mathbb{F}}(r+1)+o(1)} \qquad (t \downarrow 0),$$

up to factors polynomial in m and r. The number of row submatrices is

$$\binom{N_m}{m+r} = \exp\{h(\gamma)m + o(m)\}$$

whenever $N_m/m \to \gamma > 1$ and $r = o(m)$. If those submatrices were independent, the minimum would occur at the scale

$$\exp\left\{-\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)} m + o(m)\right\}.$$

The main theorem proves that this independence heuristic gives the correct exponential scale despite the strong dependence among the row submatrices.

1.1 Gaussian theorem

Throughout the paper, a standard real Gaussian is $N(0,1)$, and a standard complex Gaussian is $(\xi + i\eta)/\sqrt{2}$, where ξ, η are independent $N(0,1)$. Thus $\mathbb{E}|g|^2 = 1$ in both fields. Put

$$d_{\mathbb{R}} = 1, \qquad d_{\mathbb{C}} = 2, \qquad h(\gamma) = \gamma \log \gamma - (\gamma - 1)\log(\gamma - 1).$$

Theorem 1.1 (Gaussian fixed-surplus law)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let $r \geq 0$ be a fixed integer, and let $G_m \in \mathbb{F}^{N_m \times m}$ have independent standard Gaussian entries. Assume $N_m/m \to \gamma > 1$ and $N_m \geq m + r$ for all large m. Then*

$$\frac{1}{m} \log M^{\mathbb{F}}_{m,r}(G_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}.$$

Equivalently,

$$M^{\mathbb{F}}_{m,r}(G_m) = \exp\left\{-\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)} m + o_P(m)\right\}.$$

If $N_m = \gamma m + O(1)$ and r is fixed, then for every fixed $\varepsilon > 0$,

$$\mathbb{P}\left\{\left|\frac{1}{m} \log M^{\mathbb{F}}_{m,r}(G_m) + \frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}\right| > \varepsilon\right\} = O_{\gamma,\varepsilon,r,\mathbb{F}}(m^{-1}).$$

It is useful to write

$$k_m = r_m + 1.$$

Then $d_{\mathbb{F}} k_m$ is the hard-edge exponent of one

$$(m + r_m) \times m = (m + k_m - 1) \times m$$

Gaussian matrix. The proof gives the following moderate-surplus form.

Theorem 1.2 (Gaussian moderate-surplus law)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, let N_m and r_m be integer-valued sequences, and let $G_m \in \mathbb{F}^{N_m \times m}$ have independent standard Gaussian entries. Suppose*

$$\frac{N_m}{m} \to \gamma > 1, \qquad k_m = r_m + 1 \in \mathbb{N}, \qquad k_m \log m = o(m), \qquad 0 \le r_m \le N_m - m \quad \text{for all large } m.$$

Then

$$\frac{k_m}{m} \log M^{\mathbb{F}}_{m,r_m}(G_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

Remark 1.3 (Origin of the condition $k_m \log m = o(m)$)**.** Two estimates in the proof have losses of size $\exp\{O(k_m \log m)\}$. The first is the one-matrix hard-edge bound for an $(m + k_m - 1) \times m$ Gaussian matrix. The second comes from pairs of row subsets with very large overlap in the second moment argument. The condition $k_m \log m = o(m)$ makes both losses negligible on the exponential m-scale. The theorem is not intended to identify the sharp transition when k_m is of order $m/\log m$.

1.2 Radial and elliptical rows

Gaussianity is not the true source of the exponent. In the upper bound, the key property is that the normal direction to the span of $m - 1$ selected rows is Haar-distributed. This remains exactly true for independent radial row directions.

Definition 1.4 (Radial row ensemble)**.** A row vector $X \in \mathbb{F}^m$ is radial if

$$X = R\Theta,$$

where Θ is uniform on the unit sphere of $\mathbb{F}^m$, $R \ge 0$, and R is independent of Θ. A matrix has independent radial rows if its rows are independent vectors $R_i \Theta_i$ with independent Haar directions and independent nonnegative radii.

Theorem 1.5 (Rotationally invariant rows)**.** *Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $A_m \in \mathbb{F}^{N_m \times m}$ have independent rows*

$$a_i = R_i \Theta_i, \qquad 1 \le i \le N_m,$$

where the Θ_i are independent Haar unit vectors, independent of the radii $R_i > 0$. Assume that N_m and r_m are integer-valued,

$$\frac{N_m}{m} \to \gamma > 1, \qquad k_m = r_m + 1 \in \mathbb{N}, \qquad k_m \log m = o(m), \qquad 0 \le r_m \le N_m - m \quad \text{for all large } m,$$

and

$$\frac{k_m}{m} \max_{1 \le i \le N_m} |\log R_i| \xrightarrow{P} 0.$$

Then

$$\frac{k_m}{m} \log M^{\mathbb{F}}_{m,r_m}(A_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

For fixed r, it is enough to assume

$$\max_{1 \le i \le N_m} |\log R_i| = o_P(m).$$

A second extension is anisotropic Gaussianity. Let Σ_m be a deterministic positive definite $m \times m$ matrix over $\mathbb{F}$ and consider rows distributed as $g_i \Sigma_m^{1/2}$, with g_i standard Gaussian.

Theorem 1.6 (Elliptical Gaussian rows)**.** *Let $G_m \in \mathbb{F}^{N_m \times m}$ be standard Gaussian, let Σ_m be deterministic positive definite, and set*

$$A_m = G_m \Sigma_m^{1/2}.$$

Assume that N_m and r_m are integer-valued,

$$\frac{N_m}{m} \to \gamma > 1, \qquad k_m = r_m + 1 \in \mathbb{N}, \qquad k_m \log m = o(m), \qquad 0 \le r_m \le N_m - m \quad \text{for all large } m,$$

and

$$\frac{k_m}{m} \max\{|\log \lambda_{\min}(\Sigma_m)|, \, |\log \lambda_{\max}(\Sigma_m)|\} \to 0.$$

Then

$$\frac{k_m}{m} \log M^{\mathbb{F}}_{m,r_m}(A_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

1.3 Phase retrieval consequence

Let $A \in \mathbb{R}^{N \times m}$ have rows $a_1, \ldots, a_N$. The real phaseless measurement map is

$$x \mod \{\pm 1\} \longmapsto |Ax| = (|\langle a_i, x\rangle|)_{i=1}^N.$$

The complement property says that this map is injective if and only if for every partition of the rows, at least one side spans $\mathbb{R}^m$; see [2, 8, 12]. Balan and Wang introduced quantitative stability parameters controlling Lipschitz stability of phaseless inversion [3, 4]. The quantity used here is a complementary-subframe obstruction to such stability, rather than a new assertion about the full optimal lower Lipschitz constant. If $\alpha(A)$ denotes the best lower Lipschitz constant of the real phaseless map with respect to the quotient metric, then Appendix B gives the deterministic inequality

$$\alpha(A) \le \omega(A).$$

Thus a small value of $\omega(A)$ is a genuine stability obstruction. At the critical threshold this obstruction is the barely spanning complement appearing in the Balan–Wang partition minimum; with positive redundancy it remains the canonical worst-complement obstruction studied in this paper. For a general real $N \times m$ matrix, define

$$\omega(A) = \inf_{\substack{J \subset [N]:\, \mathrm{rank}(A_J) < m \\ |J^c| \ge m}} \sigma_m(A_{J^c}),$$

with the convention $\omega(A) = +\infty$ if the admissible family is empty. In the full-spark phase-retrieval regime considered below the admissible family is nonempty, so this infimum is a minimum. The condition $|J^c| \ge m$ is included only to ensure that the least singular value $\sigma_m(A_{J^c})$ is defined in the usual rectangular sense. The rectangular theorem gives the exact fixed- and moderate-redundancy law for this parameter.

Theorem 1.7 (Phase retrieval redundancy law)**.** *Let $A_m \in \mathbb{R}^{(2m-1+r_m) \times m}$ be one of the following ensembles:*

(i) standard Gaussian;

(ii) radial with independent Haar directions and radii satisfying the radial condition above;

(iii) elliptical Gaussian satisfying the covariance condition above with $N_m = 2m - 1 + r_m$.

Assume that $r_m \geq 0$ is integer-valued, $k_m = r_m + 1$, and $k_m \log m = o(m)$. Then

$$\frac{k_m}{m} \log \omega(A_m) \xrightarrow{P} -\log 4.$$

In particular, for a fixed integer $r \geq 0$,

$$\omega(A_m) = 4^{-m/(r+1) + o_P(m)}.$$

Equivalently, the sharp fixed-redundancy Gaussian base is

$$\beta_r = 4^{-1/(r+1)}.$$

Corollary 1.8 (A stability barrier for slowly growing redundancy)**.** *Under the assumptions of Theorem 1.7, if*

$$r_m + 1 = o\left(\frac{m}{\log m}\right),$$

then for every fixed $q > 0$,

$$\mathbb{P}\{\omega(A_m) \leq m^{-q}\} \to 1.$$

Consequently, the conditioning scale $1/\omega(A_m)$ is super-polynomial in m with probability tending to one.

1.4 Universality beyond rotational invariance

For non-Gaussian iid entries, the lower bound is usually easier than the upper bound. A union bound needs only a one-matrix least-singular-value estimate. The upper bound requires producing one bad submatrix among exponentially many dependent candidates. Gaussianity enters because the normal to a random span is Haar-distributed. For general iid entries, one needs a replacement: uniform delocalization of all relevant random normals and density-level small-ball estimates at exponentially small scales.

Rather than claim a universality theorem under insufficiently checked hypotheses, Section 8 proves an abstract criterion. It isolates the exact inputs needed for the second moment proof and explains why smooth continuous laws and atomic laws lead to different technical problems.

2 Preliminaries

2.1 Notation and deterministic facts

For an $N \times m$ matrix A and $T \subset [N]$, A_T denotes the row submatrix formed from rows indexed by T. If A_T has at least m rows, $\sigma_m(A_T)$ denotes its least singular value, namely the smallest singular value among the m singular values of the tall matrix A_T. We do not use $\sigma_m(A_T)$ when $|T| < m$, except through definitions that explicitly exclude that case. The scalar-field dimension is $d_{\mathbb{R}} = 1$ and $d_{\mathbb{C}} = 2$.

We use the following deterministic monotonicity several times. If B is obtained from C by deleting rows, then

$$B^* B \leq C^* C$$

as positive semidefinite matrices. Hence, by the min-max principle,

$$\sigma_m(B) \le \sigma_m(C).$$

Equivalently, adding rows cannot decrease the least singular value.

Lemma 2.1 (Binomial estimates)**.** *Let N_m and k_m be integer-valued, let $N_m/m \to \gamma > 1$, and let $k_m \log m = o(m)$. Then*

$$\log \binom{N_m}{m + k_m - 1} = h(\gamma)m + o(m).$$

If

$$L_m = \binom{N_m - k_m}{m-1},$$

then

$$\log L_m = h(\gamma)m + o(m).$$

If $N_m = \gamma m + O(1)$ and $k_m = k$ is fixed, then

$$\binom{N_m - k}{m-1} = \Theta_{\gamma,k}(m^{-1/2}e^{h(\gamma)m}).$$

Proof. We give the details because the same estimate is used on both sides of the proof. Stirling's formula in the form

$$\log \binom{n}{s} = nH(s/n) + O(\log n), \qquad H(x) = -x\log x - (1-x)\log(1-x),$$

holds uniformly for $1 \le s \le n-1$. In the first estimate, $s_m = m + k_m - 1 = m + o(m)$ because $k_m \log m = o(m)$, and $N_m/m \to \gamma$. Hence

$$\begin{aligned} N_m H(s_m/N_m) &= s_m \log \frac{N_m}{s_m} + (N_m - s_m)\log \frac{N_m}{N_m - s_m} \\ &= m\log\gamma + (\gamma-1)m\log\frac{\gamma}{\gamma-1} + o(m) \\ &= \big(\gamma\log\gamma - (\gamma-1)\log(\gamma-1)\big)m + o(m). \end{aligned}$$

The $O(\log n)$ term is $o(m)$, proving the first estimate. The proof of the second is identical with $N_m - k_m = (\gamma + o(1))m$ and $s = m-1$.

For the fixed-k estimate, use the sharper Stirling formula

$$n! = \sqrt{2\pi n}\,(n/e)^n(1 + O(n^{-1}))$$

for the three factorials in $\binom{N_m - k}{m-1}$. Since all three arguments are linear in m and their ratios converge to positive constants depending only on γ and k, the exponential term is $e^{h(\gamma)m}$ and the remaining prefactor is comparable to $m^{-1/2}$, with constants depending on γ and k. □

2.2 Hard-edge tails for one rectangular Gaussian matrix

Let $G^{\mathbb{F}}_{m,k}$ denote an $(m+k-1)\times m$ standard Gaussian matrix over $\mathbb{F}$. The least singular value of $G^{\mathbb{F}}_{m,k}$ has hard-edge exponent $d_{\mathbb{F}}k$. We need a version that is uniform enough in k to be used when $k\log m = o(m)$.

Lemma 2.2 (Rectangular hard-edge tail). *There is a constant $C_{\mathbb{F}} < \infty$ such that for all $m \geq 2$, all $1 \leq k \leq m$, and all $0 < t \leq 1$,*

$$\mathbb{P}\{\sigma_m(G^{\mathbb{F}}_{m,k}) \leq t\} \leq \exp\{C_{\mathbb{F}} k \log m\} t^{d_{\mathbb{F}} k}.$$

Moreover, if $g_1, \ldots, g_k$ are independent standard scalar Gaussians over $\mathbb{F}$ and

$$p^{\mathbb{F}}_k(t) = \mathbb{P}\left\{\sum_{j=1}^{k} |g_j|^2 \leq t^2\right\},$$

then, for $1 \leq k \leq m$ and $0 < t \leq 1$,

$$\exp\{-C_{\mathbb{F}} k \log m\} t^{d_{\mathbb{F}} k} \leq p^{\mathbb{F}}_k(t) \leq \exp\{C_{\mathbb{F}} k \log m\} t^{d_{\mathbb{F}} k}.$$

Proof. Write $\beta = d_{\mathbb{F}} \in \{1, 2\}$ and $a = \beta k/2$. Let $n = m + k - 1$. The ordered eigenvalues

$$0 < \lambda_1 < \lambda_2 < \cdots < \lambda_m$$

of $(G^{\mathbb{F}}_{m,k})^* G^{\mathbb{F}}_{m,k}$ have the Laguerre density

$$\frac{1}{Z^{\mathrm{ord}}_{m,a,\beta}} \prod_{1 \leq i < j \leq m} (\lambda_j - \lambda_i)^{\beta} \prod_{i=1}^{m} \lambda_i^{a-1} e^{-\beta \lambda_i/2}$$

on the chamber $0 < \lambda_1 < \cdots < \lambda_m$. The exponent $a - 1$ appears because $n - m + 1 = k$. The normalizing constant is the ordered Laguerre-Selberg integral

$$Z^{\mathrm{ord}}_{m,a,\beta} = \frac{1}{m!} \left(\frac{\beta}{2}\right)^{-ma - \beta m(m-1)/2} \prod_{j=1}^{m} \frac{\Gamma(1 + j\beta/2)\Gamma(a + (j-1)\beta/2)}{\Gamma(1 + \beta/2)}.$$

The formula is the standard normalization of the real and complex Wishart ensembles. We use it only through the ratio computed below.

Set $s = t^2 \leq 1$. Since $\sigma_m(G^{\mathbb{F}}_{m,k})^2 = \lambda_1$,

$$\mathbb{P}\{\sigma_m(G^{\mathbb{F}}_{m,k}) \leq t\} = \frac{1}{Z^{\mathrm{ord}}_{m,a,\beta}} \int_0^s x^{a-1} e^{-\beta x/2} I(x)\, dx,$$

where

$$I(x) = \int_{x < \lambda_2 < \cdots < \lambda_m} \prod_{j=2}^{m} (\lambda_j - x)^{\beta} \prod_{2 \leq i < j \leq m} (\lambda_j - \lambda_i)^{\beta} \times \prod_{j=2}^{m} \lambda_j^{a-1} e^{-\beta \lambda_j/2}\, d\lambda_2 \cdots d\lambda_m.$$

For $\lambda_j \geq x$, $(\lambda_j - x)^{\beta} \leq \lambda_j^{\beta}$. Enlarging the domain of integration to $0 < \lambda_2 < \cdots < \lambda_m$ gives

$$I(x) \leq Z^{\mathrm{ord}}_{m-1,a+\beta,\beta}.$$

Therefore

$$\mathbb{P}\{\sigma_m(G^{\mathbb{F}}_{m,k}) \leq t\} \leq \frac{1}{a} \frac{Z^{\mathrm{ord}}_{m-1,a+\beta,\beta}}{Z^{\mathrm{ord}}_{m,a,\beta}} t^{2a}.$$

It remains to bound the ratio. Put $b = \beta/2$, so $a = bk$. We record the cancellation explicitly. Substituting the two Laguerre-Selberg constants gives three contributions. The factorials give

$$\frac{m!}{(m-1)!} = m.$$

The powers of $\beta/2$ give

$$\left(\frac{\beta}{2}\right)^{ma+\beta m(m-1)/2-(m-1)(a+\beta)-\beta(m-1)(m-2)/2} = \left(\frac{\beta}{2}\right)^{a}.$$

The factors $\Gamma(1 + j\beta/2)$ cancel except for the denominator $\Gamma(1 + bm)$. Finally,

$$\frac{\prod_{j=1}^{m-1} \Gamma(a + \beta + (j-1)\beta/2)}{\prod_{j=1}^{m} \Gamma(a + (j-1)\beta/2)} = \frac{\prod_{j=1}^{m-1} \Gamma(b(k + j + 1))}{\prod_{j=1}^{m} \Gamma(b(k + j - 1))} = \frac{\Gamma(b(k+m))}{\Gamma(bk)\Gamma(b(k+1))}.$$

Therefore

$$\frac{Z^{\mathrm{ord}}_{m-1,a+\beta,\beta}}{Z^{\mathrm{ord}}_{m,a,\beta}} = C_\beta m \left(\frac{\beta}{2}\right)^{a} \frac{\Gamma(b(k+m))}{\Gamma(bk)\Gamma(b(k+1))\Gamma(1+bm)},$$

where C_β depends only on β. The gamma-ratio bound in Lemma A.1 gives

$$m\frac{\Gamma(b(k+m))}{\Gamma(bk)\Gamma(b(k+1))\Gamma(1+bm)} \le \exp\{C_\beta k \log m\}$$

uniformly for $1 \le k \le m$. Since $a^{-1} \le 2$ and $(\beta/2)^a \le e^{C_\beta k}$, combining these estimates yields

$$\mathbb{P}\{\sigma_m(G^{\mathbb{F}}_{m,k}) \le t\} \le \exp\{C_{\mathbb{F}} k \log m\} t^{\beta k},$$

which proves the first claim.

We now prove the estimate for $p^{\mathbb{F}}_k(t)$. The random variable

$$W = \sum_{j=1}^{k} |g_j|^2$$

has gamma density

$$f_W(u) = \frac{c^a}{\Gamma(a)} u^{a-1} e^{-cu}, \qquad u > 0, \qquad c = \frac{\beta}{2}, \qquad a = \frac{\beta k}{2}.$$

Since $t \le 1$, the integral over $0 \le u \le t^2$ satisfies

$$e^{-c}\frac{c^a t^{2a}}{\Gamma(a+1)} \le p^{\mathbb{F}}_k(t) \le \frac{c^a t^{2a}}{\Gamma(a+1)}.$$

Finally, Stirling's formula gives

$$\exp\{-C_{\mathbb{F}} k \log m\} \le \frac{c^a}{\Gamma(a+1)} \le \exp\{C_{\mathbb{F}} k \log m\}$$

for $1 \le k \le m$. Because $2a = \beta k = d_{\mathbb{F}} k$, the result follows. □

3 The lower estimate

The lower estimate says that, with high probability, no $(m+r_m)$-row submatrix has least singular value below the predicted scale. It is a union bound using Lemma 2.2.

Proposition 3.1 (Gaussian lower estimate)**.** *Assume the hypotheses of Theorem 1.2. For every fixed $\varepsilon > 0$,*

$$\mathbb{P}\left\{M^{\mathbb{F}}_{m,r_m}(G_m) \le \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}} + \varepsilon\right)\frac{m}{k_m}\right]\right\} \to 0.$$

Equivalently,

$$\liminf_{m\to\infty} \frac{k_m}{m}\log M^{\mathbb{F}}_{m,r_m}(G_m) \ge -\frac{h(\gamma)}{d_{\mathbb{F}}}$$

in probability. If $k_m = k$ is fixed and $N_m = \gamma m + O(1)$, then the probability above is exponentially small in m.

Proof. Set

$$t_m = \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}} + \varepsilon\right)\frac{m}{k_m}\right].$$

For a fixed subset $T \subset [N_m]$ with

$$|T| = m + r_m = m + k_m - 1,$$

the matrix $(G_m)_T$ has the same distribution as $G^{\mathbb{F}}_{m,k_m}$. Since $k_m \log m = o(m)$, we have $k_m \le m$ for all large m, so Lemma 2.2 applies. By the union bound,

$$\begin{aligned}
\mathbb{P}\{M^{\mathbb{F}}_{m,r_m}(G_m) \le t_m\} &\le \binom{N_m}{m+k_m-1}\mathbb{P}\{\sigma_m(G^{\mathbb{F}}_{m,k_m}) \le t_m\} \\
&\le \exp\{h(\gamma)m + o(m)\}\exp\{C_{\mathbb{F}} k_m \log m\} t_m^{d_{\mathbb{F}} k_m} \\
&= \exp\{-d_{\mathbb{F}}\varepsilon m + o(m) + C_{\mathbb{F}} k_m \log m\}.
\end{aligned}$$

The last exponent tends to $-\infty$ because $k_m \log m = o(m)$. This proves the first claim.

If $k_m = k$ is fixed and $N_m = \gamma m + O(1)$, Lemma 2.1 gives a polynomial prefactor times $e^{h(\gamma)m}$, while Lemma 2.2 contributes only a polynomial factor in m. Hence the preceding bound is $\exp\{-c_{\varepsilon,\mathbb{F}} m + O(\log m)\}$ for some $c_{\varepsilon,\mathbb{F}} > 0$. □

4 The upper estimate

The upper estimate is the heart of the argument. We mark k_m rows and look for a set of $m-1$ additional rows whose span is almost orthogonal to all marked rows. If such a span exists, then the union of those $m-1$ rows with the k_m marked rows gives an $(m+r_m)\times m$ submatrix with a small least singular value.

Throughout this section we suppress the subscript m and write

$$N = N_m, \qquad k = k_m, \qquad r = r_m,$$

except where doing so could cause confusion.

4.1 The marked-row construction

Write the last k rows of G_m as

$$y_1, \dots, y_k \in \mathbb{F}^m,$$

and call them the marked rows. Write the remaining $N-k$ rows as

$$b_1, \dots, b_{N-k}.$$

Let

$$\mathcal{S} = \{S \subset [N-k] : |S| = m-1\}, \qquad L = |\mathcal{S}| = \binom{N-k}{m-1}.$$

For $S \in \mathcal{S}$, define

$$H_S = \operatorname{span}_{\mathbb{F}}\{b_i : i \in S\}.$$

Almost surely H_S is a hyperplane. Choose a unit normal u_S to H_S; in the complex case this normal is defined only up to multiplication by a scalar of modulus one, but all events below are phase-invariant. For $t > 0$, define

$$X_S(t) = \mathbf{1}\left\{\sum_{j=1}^{k} |\langle y_j, u_S\rangle|^2 \le t^2\right\}, \qquad Z_t = \sum_{S \in \mathcal{S}} X_S(t).$$

Lemma 4.1 (Marked rows imply a bad submatrix)**.** *If $Z_t > 0$, then*

$$M^{\mathbb{F}}_{m,k-1}(G_m) \le t.$$

Proof. If $Z_t > 0$, then there exists $S \in \mathcal{S}$ such that $X_S(t) = 1$. Let T be the union of S and the k marked rows:

$$T = S \cup \{N-k+1, \dots, N\}.$$

Then

$$|T| = (m-1) + k = m + k - 1 = m + r.$$

Since u_S is a unit vector orthogonal to the $m-1$ unmarked rows indexed by S,

$$\|(G_m)_T u_S\|_2^2 = \sum_{i \in S} |\langle b_i, u_S\rangle|^2 + \sum_{j=1}^{k} |\langle y_j, u_S\rangle|^2 = \sum_{j=1}^{k} |\langle y_j, u_S\rangle|^2 \le t^2.$$

The variational characterization of the least singular value gives

$$\sigma_m((G_m)_T) \le \|(G_m)_T u_S\|_2 \le t.$$

Taking the minimum over all T proves the claim. □

Because the marked rows are independent of u_S and are standard Gaussian, for every fixed unit vector u the scalars

$$\langle y_1, u\rangle, \dots, \langle y_k, u\rangle$$

are independent standard scalar Gaussians over $\mathbb{F}$. Therefore

$$\mathbb{E} Z_t = L p_k^{\mathbb{F}}(t),$$

where $p_k^{\mathbb{F}}$ is defined in Lemma 2.2. The main task is to prove that Z_t is nonzero with high probability at the predicted scale.

4.2 Geometry of two random hyperplanes

For $S, T \in \mathcal{S}$, define
$$D(S,T) = m - |S \cap T|.$$
If $S = T$, then $D(S,T) = 1$. If $S \neq T$, then $D(S,T) \geq 2$, because two distinct subsets of the same cardinality cannot differ in only one element.

Lemma 4.2 (Normals for overlapping row sets)**.** *Fix distinct $S, T \in \mathcal{S}$ and set $D = D(S,T)$. Conditional on the rows indexed by $S \cap T$, the two normal lines $\mathbb{F}u_S$ and $\mathbb{F}u_T$ are independent Haar-distributed lines in the D-dimensional space*
$$E^{\perp}, \qquad E = \operatorname{span}_{\mathbb{F}}\{b_i : i \in S \cap T\}.$$

Proof. Almost surely the vectors indexed by $S \cap T$ are linearly independent, so
$$\dim_{\mathbb{F}} E = |S \cap T| = m - D, \qquad \dim_{\mathbb{F}} E^{\perp} = D.$$
The set $S \setminus T$ has cardinality $D - 1$. Conditional on E, the orthogonal projections of the rows $\{b_i : i \in S \setminus T\}$ onto $E^{\perp}$ are independent standard Gaussian vectors in $E^{\perp}$. This follows from the rotational invariance of the Gaussian distribution and from the independence of the rows outside $S \cap T$. Their span is almost surely a $(D-1)$-dimensional hyperplane in $E^{\perp}$, and its normal line is Haar-distributed on the projective space of lines in $E^{\perp}$. The same argument applies to the rows indexed by $T \setminus S$.

The two families of projected rows, those from $S \setminus T$ and those from $T \setminus S$, are conditionally independent given the rows in $S \cap T$. Hence the two normal lines are conditionally independent Haar lines in $E^{\perp}$. $\square$

For $D \geq 2$, define
$$q^{\mathbb{F}}_{D,k}(t) = \mathbb{P}\left\{ \sum_{j=1}^{k} |\langle g_j, u\rangle|^2 \leq t^2,\ \sum_{j=1}^{k} |\langle g_j, v\rangle|^2 \leq t^2 \right\},$$
where u, v are independent Haar unit vectors in $\mathbb{F}^D$ and $g_1, \ldots, g_k$ are independent standard Gaussian vectors in $\mathbb{F}^D$, independent of u, v.

Lemma 4.3 (Two-slab estimate for k marked rows)**.** *There is a constant $C_{\mathbb{F}} < \infty$ such that for all $1 \leq k \leq m$, all $0 < t \leq 1$, and all $D \geq 4k + 4$,*
$$q^{\mathbb{F}}_{D,k}(t) \leq \left(p^{\mathbb{F}}_k(t)\right)^2 \exp\left\{ C_{\mathbb{F}} \frac{k}{D} + C_{\mathbb{F}} k t^2 \right\}.$$
For all $D \geq 2$,
$$q^{\mathbb{F}}_{D,k}(t) \leq p^{\mathbb{F}}_k(t).$$

Proof. The crude bound follows because the joint event is contained in either one of the two marginal events, each of which has probability $p^{\mathbb{F}}_k(t)$.

We prove the sharper estimate. Fix u, v and put
$$\rho = \langle u, v\rangle, \qquad s^2 = 1 - |\rho|^2.$$
For one marked row g, the pair
$$(\langle g, u\rangle, \langle g, v\rangle)$$

is a centered Gaussian vector over $\mathbb{F}^2$ with covariance matrix

$$\begin{pmatrix} 1 & \rho \\ \bar{\rho} & 1 \end{pmatrix},$$

whose determinant is s^2. For k independent marked rows, let

$$X = (\langle g_j, u\rangle)_{j=1}^k, \qquad Y = (\langle g_j, v\rangle)_{j=1}^k.$$

Then $(X, Y) \in \mathbb{F}^k \times \mathbb{F}^k$ has joint density

$$f_\rho(x, y) = s^{-d_{\mathbb{F}}k}\phi_k(x)\phi_k\left(\frac{y - \rho x}{s}\right),$$

where ϕ_k is the standard Gaussian density on $\mathbb{F}^k$. In particular, the density is maximized at $(0, 0)$ and

$$f_\rho(x, y) \le s^{-d_{\mathbb{F}}k}\phi_k(0)^2$$

for all x, y. Let $B_t = \{z \in \mathbb{F}^k : \|z\|_2 \le t\}$. Conditional on u, v,

$$\mathbb{P}\{X \in B_t,\, Y \in B_t \mid u, v\} \le s^{-d_{\mathbb{F}}k}\phi_k(0)^2 \operatorname{vol}(B_t)^2.$$

On the other hand, for $z \in B_t$ and $t \le 1$,

$$\phi_k(z) \ge e^{-C_{\mathbb{F}}t^2}\phi_k(0).$$

Therefore

$$p_k^{\mathbb{F}}(t) = \int_{B_t} \phi_k(z)\, dz \ge e^{-C_{\mathbb{F}}t^2}\phi_k(0)\operatorname{vol}(B_t).$$

Combining these estimates gives

$$\mathbb{P}\{X \in B_t,\, Y \in B_t \mid u, v\} \le \left(p_k^{\mathbb{F}}(t)\right)^2 e^{C_{\mathbb{F}}t^2}(1 - |\rho|^2)^{-d_{\mathbb{F}}k/2}.$$

We may replace $e^{C_{\mathbb{F}}t^2}$ by $e^{C_{\mathbb{F}}kt^2}$, which is weaker but convenient.

It remains to average over u, v. The random variable

$$R = |\langle u, v\rangle|^2$$

has the beta distribution $\mathrm{Beta}(1/2, (D-1)/2)$ over $\mathbb{R}$ and $\mathrm{Beta}(1, D-1)$ over $\mathbb{C}$. This is the usual distribution of the squared first coordinate of a Haar unit vector after rotating u to the first coordinate direction.

Set

$$\alpha = \frac{d_{\mathbb{F}}k}{2}.$$

In the real case, $(a_0, b_0) = (1/2, (D-1)/2)$ and $\alpha = k/2$. In the complex case, $(a_0, b_0) = (1, D-1)$ and $\alpha = k$. In both cases $D \ge 4k + 4$ implies $0 \le \alpha \le b_0/2$. The beta integral gives

$$\mathbb{E}(1 - R)^{-\alpha} = \frac{B(a_0, b_0 - \alpha)}{B(a_0, b_0)} = \frac{\Gamma(b_0 - \alpha)\Gamma(a_0 + b_0)}{\Gamma(b_0)\Gamma(a_0 + b_0 - \alpha)}.$$

Equivalently, taking logarithms and using $\psi = \Gamma'/\Gamma$,

$$\log \mathbb{E}(1 - R)^{-\alpha} = \int_0^\alpha \left[\psi(a_0 + b_0 - s) - \psi(b_0 - s)\right] ds.$$

For $0 \le s \le \alpha$, the lower argument satisfies $b_0 - s \ge b_0/2 \asymp D$. The estimate

$$0 \le \psi(x + a_0) - \psi(x) \le \frac{C_{\mathbb{F}}}{x}, \qquad x \ge 1,$$

which follows from the digamma series or from Lemma A.2, gives

$$\log \mathbb{E}(1-R)^{-\alpha} \le C_{\mathbb{F}} \frac{\alpha}{D} \le C_{\mathbb{F}} \frac{k}{D}.$$

Thus

$$\mathbb{E}(1 - |\langle u, v\rangle|^2)^{-d_{\mathbb{F}} k/2} \le \exp\{C_{\mathbb{F}} k/D\}.$$

Averaging the conditional estimate proves the claim. □

4.3 Overlap combinatorics

Let S, T be independent uniform elements of $\mathcal{S}$. The random variable $D = D(S,T) = m - |S \cap T|$ is hypergeometric.

Lemma 4.4 (Overlap estimates)**.** *Let $n = N - k$ and let $\mathcal{S} = \{S \subset [n] : |S| = m-1\}$. If S, T are independent uniform elements of $\mathcal{S}$ and $D = m - |S \cap T|$, then*

$$\mathbb{P}\{D = d\} = \frac{\binom{m-1}{d-1}\binom{N-k-m+1}{d-1}}{\binom{N-k}{m-1}},$$

with the convention that impossible binomial coefficients are zero. Moreover, if $N/m \to \gamma > 1$ and $k = o(m)$, then

$$\mathbb{E}\frac{1}{D} = O_\gamma(m^{-1}).$$

Finally, for every integer $a \ge 1$,

$$\mathbb{P}\{D \le a\} \le \exp\{C_\gamma a \log m\} \binom{N-k}{m-1}^{-1}$$

for all large m.

Proof. Fix S. To have $D = d$, the set T must remove $d-1$ elements from S and add $d-1$ elements from the complement of S in $[N-k]$, which has size $N-k-m+1$. Since the total number of choices for T is $\binom{N-k}{m-1}$, this proves the formula for $\mathbb{P}\{D = d\}$.

For the expectation, put $n = N-k$ and $s = m-1$. Write $j = d-1$. Using the preceding formula,

$$\mathbb{E}\frac{1}{D} = \frac{1}{\binom{n}{s}} \sum_{j \ge 0} \frac{1}{j+1} \binom{s}{j} \binom{n-s}{j}.$$

The identity

$$\frac{1}{j+1}\binom{s}{j} = \frac{1}{s+1}\binom{s+1}{j+1}$$

turns this into

$$\mathbb{E}\frac{1}{D} = \frac{1}{\binom{n}{s}} \frac{1}{s+1} \sum_{j \ge 0} \binom{s+1}{j+1} \binom{n-s}{j}.$$

By Vandermonde's identity, the sum equals $\binom{n+1}{s}$. Therefore

$$\mathbb{E}\frac{1}{D} = \frac{1}{s+1}\frac{\binom{n+1}{s}}{\binom{n}{s}} = \frac{n+1}{(s+1)(n+1-s)} = \frac{n+1}{m(n-m+2)}.$$

Since $n = N - k = (\gamma + o(1))m$, the last expression is $O_\gamma(m^{-1})$.

For the small-overlap bound, sum the probability formula over $1 \le d \le a$. For each such d,

$$\binom{m-1}{d-1}\binom{N-k-m+1}{d-1} \le (Cm)^{2(d-1)} \le \exp\{Ca\log m\},$$

because $N = O_\gamma(m)$ and $k = o(m)$. Multiplying by the at most a possible values of d only changes the constant. $\square$

4.4 The second moment

Proposition 4.5 (Gaussian upper estimate)**.** *Assume the hypotheses of Theorem 1.2. For every fixed $\varepsilon > 0$,*

$$\mathbb{P}\left\{M^{\mathbb{F}}_{m,r_m}(G_m) > \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}} - \varepsilon\right)\frac{m}{k_m}\right]\right\} \to 0.$$

Equivalently,

$$\limsup_{m\to\infty} \frac{k_m}{m}\log M^{\mathbb{F}}_{m,r_m}(G_m) \le -\frac{h(\gamma)}{d_{\mathbb{F}}}$$

in probability. If $k_m = k$ is fixed and $N_m = \gamma m + O(1)$, then the probability above is $O_{\gamma,\varepsilon,k,\mathbb{F}}(m^{-1})$.

Proof. It is enough to prove the statement for

$$0 < \varepsilon < \frac{h(\gamma)}{2d_{\mathbb{F}}};$$

larger ε follow by replacing ε with a smaller positive number. Set

$$t_m = \exp\left[-\left(\frac{h(\gamma)}{d_{\mathbb{F}}} - \varepsilon\right)\frac{m}{k_m}\right].$$

Then $0 < t_m \le 1$ for all large m. Since $k_m \log m = o(m)$, for every fixed $A > 0$ we have $t_m \le m^{-A}$ for all large m, and in particular

$$k_m t_m^2 \to 0.$$

Let Z_{t_m} be the marked-row count from the previous subsection. Its mean is

$$\mu_m = \mathbb{E}Z_{t_m} = \binom{N_m - k_m}{m-1} p^{\mathbb{F}}_{k_m}(t_m).$$

By Lemmas 2.1 and 2.2,

$$\begin{aligned}\mu_m &\ge \exp\{h(\gamma)m + o(m) - C_{\mathbb{F}}k_m\log m\}t_m^{d_{\mathbb{F}}k_m}\\ &= \exp\{d_{\mathbb{F}}\varepsilon m + o(m) - C_{\mathbb{F}}k_m\log m\} \to \infty.\end{aligned}$$

We prove that $\mathrm{Var}(Z_{t_m}) = o(\mu_m^2)$. For notational clarity, write $t = t_m$ and $k = k_m$ until the end of the proof. The diagonal contribution satisfies

$$\frac{1}{\mu_m^2}\sum_{S\in\mathcal{S}}\mathbb{E}X_S(t)^2 = \frac{Lp^{\mathbb{F}}_k(t)}{(Lp^{\mathbb{F}}_k(t))^2} = \frac{1}{\mu_m} = o(1),$$

where $X_S(t)^2 = X_S(t)$ was used.

For $S \neq T$, Lemma 4.2 and the independence of the marked rows from the unmarked rows give

$$\mathbb{E} X_S(t) X_T(t) = q^{\mathbb{F}}_{D(S,T),k}(t).$$

Consequently, if S, T are independent uniform elements of $\mathcal{S}$, then

$$\frac{\mathbb{E} Z_t^2}{\mu_m^2} \leq \frac{1}{\mu_m} + \mathbb{E}\left[\frac{q^{\mathbb{F}}_{D,k}(t)}{(p^{\mathbb{F}}_k(t))^2} \mathbf{1}_{\{D \geq 2\}}\right].$$

We split the expectation according to whether $D \geq 4k+4$.

On the good-overlap event $D \geq 4k+4$, Lemma 4.3 gives

$$\frac{q^{\mathbb{F}}_{D,k}(t)}{(p^{\mathbb{F}}_k(t))^2} \leq \exp\left\{C_{\mathbb{F}} \frac{k}{D} + C_{\mathbb{F}} k t^2\right\}.$$

Since $D \geq 4k+4$, the exponent $C_{\mathbb{F}} k/D$ is bounded by an absolute constant. Hence $e^x \leq 1 + C' x$ on the relevant bounded interval, and Lemma 4.4 gives

$$\begin{aligned}
\mathbb{E}\left[\exp\left\{C_{\mathbb{F}} \frac{k}{D} + C_{\mathbb{F}} k t^2\right\} \mathbf{1}_{\{D \geq 4k+4\}}\right] &\leq e^{C_{\mathbb{F}} k t^2} \left(1 + C'_{\mathbb{F}} k \mathbb{E} \frac{1}{D}\right) \\
&\leq (1 + o(1))(1 + O_\gamma(k/m)) \\
&= 1 + o(1),
\end{aligned}$$

because $k = o(m)$ and $kt^2 \to 0$.

On the bad-overlap event $2 \leq D < 4k+4$, the crude bound in Lemma 4.3 gives

$$\frac{q^{\mathbb{F}}_{D,k}(t)}{(p^{\mathbb{F}}_k(t))^2} \leq \frac{1}{p^{\mathbb{F}}_k(t)}.$$

Using Lemmas 2.2 and 4.4,

$$\begin{aligned}
\mathbb{E}\left[\frac{q^{\mathbb{F}}_{D,k}(t)}{(p^{\mathbb{F}}_k(t))^2} \mathbf{1}_{\{2 \leq D < 4k+4\}}\right] &\leq \mathbb{P}\{D < 4k+4\} \frac{1}{p^{\mathbb{F}}_k(t)} \\
&\leq \exp\{Ck \log m\} L^{-1} \exp\{C_{\mathbb{F}} k \log m\} t^{-d_{\mathbb{F}} k} \\
&\leq \exp\{O(k \log m) - h(\gamma) m + o(m) + (h(\gamma) - d_{\mathbb{F}} \varepsilon) m\} \\
&= \exp\{O(k \log m) - d_{\mathbb{F}} \varepsilon m + o(m)\} \to 0.
\end{aligned}$$

Combining the diagonal, good-overlap, and bad-overlap estimates, we obtain

$$\frac{\mathbb{E} Z_t^2}{\mu_m^2} \leq 1 + o(1).$$

Thus $\mathrm{Var}(Z_t) = o(\mu_m^2)$, and Chebyshev's inequality gives

$$\mathbb{P}\{Z_t = 0\} \leq \frac{\mathrm{Var}(Z_t)}{\mu_m^2} \to 0.$$

By Lemma 4.1, $Z_t > 0$ implies $M^{\mathbb{F}}_{m,r_m}(G_m) \leq t$, which proves the main claim.

Now assume $k_m = k$ is fixed and $N_m = \gamma m + O(1)$. The diagonal term $1/\mu_m$ is exponentially small. The bad-overlap term is also exponentially small. In the good-overlap term, Lemma 4.4 gives $\mathbb{E}(1/D) = O_\gamma(1/m)$, while kt_m^2 is exponentially small. Therefore

$$\frac{\mathbb{E}Z_t^2}{\mu_m^2} \le 1 + O_{\gamma,\varepsilon,k,\mathbb{F}}(m^{-1}).$$

Chebyshev's inequality yields $\mathbb{P}\{Z_t = 0\} = O_{\gamma,\varepsilon,k,\mathbb{F}}(m^{-1})$, and the deterministic implication in Lemma 4.1 gives the claimed rate. □

5 Proof of the Gaussian theorems

Proof of Theorem 1.2. Proposition 3.1 gives, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\left\{\frac{k_m}{m}\log M^{\mathbb{F}}_{m,r_m}(G_m) < -\frac{h(\gamma)}{d_{\mathbb{F}}} - \varepsilon\right\} \to 0.$$

Proposition 4.5 gives, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\left\{\frac{k_m}{m}\log M^{\mathbb{F}}_{m,r_m}(G_m) > -\frac{h(\gamma)}{d_{\mathbb{F}}} + \varepsilon\right\} \to 0.$$

The two estimates are exactly convergence in probability to $-h(\gamma)/d_{\mathbb{F}}$. □

Proof of Theorem 1.1. Take $k_m = r + 1$ fixed in Theorem 1.2. Then

$$\frac{r+1}{m}\log M^{\mathbb{F}}_{m,r}(G_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}},$$

which is equivalent to the fixed-surplus law. The rate statement follows by combining the exponential lower-tail estimate in Proposition 3.1 with the $O(m^{-1})$ upper-tail estimate in Proposition 4.5. □

Remark 5.1 (The role of $r+1$)**.** The number $r + 1$ has two equivalent interpretations. It is the hard-edge parameter k of a single

$$(m + r) \times m = (m + k - 1) \times m$$

Gaussian matrix. It is also the number of marked rows that must be simultaneously small in the normal direction to the span of $m - 1$ unmarked rows. Each extra row adds one scalar hard-edge constraint over $\mathbb{R}$ and two real scalar constraints over $\mathbb{C}$, which is why the exponent is divided by $d_{\mathbb{F}}(r + 1)$.

6 Radial and elliptical ensembles

This section proves Theorems 1.5 and 1.6. The proofs are comparison arguments. Once the Gaussian theorem is known, row normalization, independent radial rescaling, and deterministic covariance deformation do not change the exponential rate unless they introduce exponential scales comparable to the hard-edge scale.

6.1 Uniform spherical directions

Let $\Theta_m \in \mathbb{F}^{N_m \times m}$ have independent Haar unit-vector rows. Let G_m be a standard Gaussian matrix. We couple G_m and Θ_m row by row by writing

$$g_i = \rho_i \theta_i, \qquad \rho_i = \|g_i\|_2, \qquad \theta_i = \frac{g_i}{\|g_i\|_2}.$$

Then the θ_i are independent Haar unit vectors and are independent of the lengths ρ_i.

Lemma 6.1 (Gaussian lengths are subexponential)**.** *Assume $N_m = O(m)$ and let $\rho_i = \|g_i\|_2$ be the row lengths of independent standard Gaussian rows in $\mathbb{F}^m$. If $k_m \log m = o(m)$, then*

$$\frac{k_m}{m} \max_{1 \le i \le N_m} |\log \rho_i| \xrightarrow{P} 0.$$

Proof. Fix $\eta > 0$ and set

$$R_m = e^{\eta m / k_m}, \qquad r_m = e^{-\eta m / k_m}.$$

Because $k_m \log m = o(m)$, R_m eventually dominates every fixed power of m, while r_m is smaller than every negative power of m.

For the upper tail, ρ_i^2 has a chi-square/gamma tail. There are constants $c, C > 0$ depending only on $\mathbb{F}$ such that, for $R \ge C\sqrt{m}$,

$$\mathbb{P}\{\rho_i \ge R\} \le e^{-cR^2}.$$

Hence

$$\mathbb{P}\left\{\max_{i \le N_m} \rho_i \ge R_m\right\} \le N_m e^{-cR_m^2} \to 0.$$

For the lower tail, the density of ρ_i is bounded near zero by $C^m s^{d_{\mathbb{F}} m - 1}$. Thus, for $0 < r \le 1$,

$$\mathbb{P}\{\rho_i \le r\} \le C^m r^{d_{\mathbb{F}} m}.$$

Therefore

$$\mathbb{P}\left\{\min_{i \le N_m} \rho_i \le r_m\right\} \le N_m \exp\{Cm - d_{\mathbb{F}} \eta m^2 / k_m\} \to 0,$$

because $m/k_m \to \infty$. The two tail bounds imply the claim. $\square$

Proposition 6.2 (Spherical rows)**.** *Let $\Theta_m \in \mathbb{F}^{N_m \times m}$ have independent Haar unit-vector rows. Under the hypotheses of Theorem 1.2,*

$$\frac{k_m}{m} \log M^{\mathbb{F}}_{m, r_m}(\Theta_m) \xrightarrow{P} -\frac{h(\gamma)}{d_{\mathbb{F}}}.$$

Proof. Use the coupling $G_m = D_\rho \Theta_m$, where $D_\rho = \mathrm{diag}(\rho_1, \ldots, \rho_{N_m})$. For every $T \subset [N_m]$ with $|T| = m + r_m$,

$$\left(\min_{i \in T} \rho_i\right) \sigma_m((\Theta_m)_T) \le \sigma_m((G_m)_T) \le \left(\max_{i \in T} \rho_i\right) \sigma_m((\Theta_m)_T).$$

Taking the minimum over T gives

$$\frac{M^{\mathbb{F}}_{m, r_m}(G_m)}{\max_i \rho_i} \le M^{\mathbb{F}}_{m, r_m}(\Theta_m) \le \frac{M^{\mathbb{F}}_{m, r_m}(G_m)}{\min_i \rho_i}.$$

After taking logarithms and multiplying by k_m/m, the row-length terms vanish in probability by Lemma 6.1. The Gaussian limit in Theorem 1.2 proves the claim. $\square$

6.2 Proof of the radial theorem

Proof of Theorem 1.5. Write

$$A_m = D_R\Theta_m, \qquad D_R = \operatorname{diag}(R_1, \dots, R_{N_m}),$$

where Θ_m has independent Haar unit-vector rows. For every T,

$$\left(\min_{i\in T} R_i\right)\sigma_m((\Theta_m)_T) \le \sigma_m((A_m)_T) \le \left(\max_{i\in T} R_i\right)\sigma_m((\Theta_m)_T).$$

Consequently, with

$$\eta_m = \max_{1\le i\le N_m} |\log R_i|,$$

we have

$$e^{-\eta_m} M^{\mathbb{F}}_{m,r_m}(\Theta_m) \le M^{\mathbb{F}}_{m,r_m}(A_m) \le e^{\eta_m} M^{\mathbb{F}}_{m,r_m}(\Theta_m).$$

Multiplying logarithms by k_m/m, the error $(k_m/m)\eta_m$ tends to zero in probability by the radial assumption. Proposition 6.2 then gives the theorem. When r is fixed, $k_m = r+1$ is fixed, and the fixed-r assumption is exactly the condition that $(k_m/m)\eta_m \to 0$ in probability. □

6.3 Proof of the elliptical theorem

Proof of Theorem 1.6. Let B be any row submatrix of G_m. The singular-value inequalities for products give

$$\sqrt{\lambda_{\min}(\Sigma_m)}\,\sigma_m(B) \le \sigma_m(B\Sigma_m^{1/2}) \le \sqrt{\lambda_{\max}(\Sigma_m)}\,\sigma_m(B).$$

Applying this to every row subset T with $|T| = m + r_m$ and taking the minimum over T yields

$$\sqrt{\lambda_{\min}(\Sigma_m)}\,M^{\mathbb{F}}_{m,r_m}(G_m) \le M^{\mathbb{F}}_{m,r_m}(A_m) \le \sqrt{\lambda_{\max}(\Sigma_m)}\,M^{\mathbb{F}}_{m,r_m}(G_m).$$

Multiplying logarithms by k_m/m, the covariance terms vanish by the covariance assumption. Theorem 1.2 proves the theorem. □

Remark 6.3 (When anisotropy changes the exponent)**.** If $\lambda_{\min}(\Sigma_m)$ or $\lambda_{\max}(\Sigma_m)$ is exponentially small or large at the scale m/k_m, then the exponent can shift. Theorem 1.6 isolates the invariant regime: covariance deformations with subexponential extreme eigenvalues at the relevant scale do not affect the hard-edge base.

7 Phase retrieval stability

We now prove the phase retrieval consequences. The key deterministic observation is that, for full-spark real frames at $N = 2m-1+r$, the complementary-subframe obstruction $\omega(A)$ is exactly the rectangular row-submatrix minimum $M^{\mathbb{R}}_{m,r}$.

Definition 7.1 (Full spark)**.** A real $N\times m$ matrix A is full spark if every m rows are linearly independent.

Gaussian, radial continuous with positive radii, and elliptical Gaussian ensembles are full spark almost surely. Indeed, for each fixed m-row subset the determinant of the corresponding $m\times m$ minor is a nonzero polynomial in the row variables, hence vanishes on a set of measure zero; a finite union over row subsets preserves probability zero.

Lemma 7.2 (Reduction of ω to $M_{m,r}$)**.** *Let $A \in \mathbb{R}^{(2m-1+r)\times m}$ be full spark, with $r \geq 0$ an integer. Then*

$$\omega(A) = M^{\mathbb{R}}_{m,r}(A).$$

The same identity holds when $r = r_m$ varies with m.

Proof. By full spark, $\operatorname{rank}(A_J) < m$ if and only if $|J| \leq m-1$. For such J, the complement has size

$$|J^c| \geq (2m-1+r) - (m-1) = m+r \geq m,$$

so the admissibility condition $|J^c| \geq m$ in the definition of ω is automatic. Therefore

$$\omega(A) = \min_{|J|\leq m-1} \sigma_m(A_{J^c}).$$

Suppose $J_1 \subset J_2$. Then $J_2^c \subset J_1^c$, so $A_{J_2^c}$ is obtained from $A_{J_1^c}$ by deleting rows. By monotonicity,

$$\sigma_m(A_{J_2^c}) \leq \sigma_m(A_{J_1^c}).$$

Hence the minimum is achieved among maximal rank-deficient sets, namely those with $|J| = m-1$. If $|J| = m-1$, then

$$|J^c| = (2m-1+r) - (m-1) = m+r.$$

Thus

$$\omega(A) = \min_{|T|=m+r} \sigma_m(A_T) = M^{\mathbb{R}}_{m,r}(A).$$

□

Proof of Theorem 1.7. For each of the listed ensembles, A_m is full spark almost surely. Lemma 7.2 gives

$$\omega(A_m) = M^{\mathbb{R}}_{m,r_m}(A_m).$$

Here

$$N_m = 2m-1+r_m = m + (m+r_m-1) = 2m + k_m - 2.$$

Since $k_m \log m = o(m)$, we have $k_m = o(m)$, and therefore

$$\frac{N_m}{m} \to 2.$$

The entropy function is continuous on $(1,\infty)$, and

$$h(2) = 2\log 2 - \log 1 = \log 4.$$

Applying Theorem 1.2, Theorem 1.5, or Theorem 1.6, according to the ensemble, gives

$$\frac{k_m}{m}\log \omega(A_m) = \frac{k_m}{m}\log M^{\mathbb{R}}_{m,r_m}(A_m) \xrightarrow{P} -\log 4.$$

For fixed r, $k_m = r+1$, and the fixed-redundancy statement follows by dividing the exponent by $r+1$. □

Proof of Corollary 1.8. Let $q > 0$ be fixed. By Theorem 1.7, for every $0 < \delta < 1$,

$$\omega(A_m) \leq \exp\left\{-(1-\delta)(\log 4)\frac{m}{k_m}\right\}$$

with probability tending to one. If $k_m = o(m/\log m)$, then $m/k_m \gg \log m$. Hence for all sufficiently large m,

$$\exp\left\{-(1-\delta)(\log 4)\frac{m}{k_m}\right\} \leq m^{-q}.$$

Combining this deterministic comparison with the preceding probability bound proves the claim. □

7.1 Interpretation for reconstruction stability

The parameter $\omega(A)$ should be read as a Balan–Wang type obstruction, not as a claim to equal the full optimal lower Lipschitz constant in every redundant regime. Let

$$\alpha(A) = \inf_{x \not\equiv \pm y} \frac{\| |Ax| - |Ay| \|_2}{\min\{\|x-y\|_2 \,, \|x+y\|_2\}}$$

be the lower Lipschitz constant of the real phaseless measurement map on $\mathbb{R}^m/\{\pm 1\}$. Appendix B proves the deterministic inequality

$$\alpha(A) \le \omega(A).$$

Consequently, the asymptotic law for $\omega(A)$ gives a rigorous upper bound on the possible lower stability scale of phaseless reconstruction.

At the critical threshold $N = 2m-1$, every partition has a side of size at most $m-1$. For a full-spark frame, that side is rank deficient, so the obstruction $\omega(A)$ is exactly the barely spanning complement obstruction in the Balan–Wang partition picture. For $N = 2m-1+r$ with $r > 0$, other partitions may have both sides spanning; the theorem here does not assert that $\omega(A)$ is always the full partition minimum. It identifies the precise exponential scale of the rank-deficient-side obstruction, which is enough to show severe ill-conditioning.

At the critical threshold $N = 2m-1$,

$$\omega(A_m) = 4^{-m+o_P(m)}.$$

With one extra measurement, $N = 2m$,

$$\omega(A_m) = 2^{-m+o_P(m)}.$$

With fixed r extra measurements,

$$\omega(A_m) = \left(4^{-1/(r+1)}\right)^m e^{o_P(m)}.$$

Thus fixed redundancy improves the exponential base exactly but does not remove exponential ill-conditioning. For growing r_m, the relevant exponent is $m/(r_m+1)$. Corollary 1.8 proves that if $r_m + 1 = o(m/\log m)$, then this stability obstruction remains super-polynomial.

8 Universality beyond rotational invariance

This section is a criterion and roadmap, not a proof of full iid universality. The Gaussian proof has two separable parts. The lower bound uses one-matrix least-singular-value tails and a union bound. The upper bound uses a second moment over random hyperplanes. Rotational invariance makes those hyperplanes Haar-distributed. For iid non-Gaussian entries, the hyperplanes are not Haar, but the same argument works if one can supply uniform one-normal and two-normal small-ball estimates at the required exponential scale.

8.1 A second-moment criterion

We state the criterion in the real case. The complex version is obtained by replacing each power t^{k_m} by t^{2k_m} and by using complex normal lines.

Let $A_m \in \mathbb{R}^{N_m \times m}$ have independent rows, $N_m/m \to \gamma > 1$, and $k_m = r_m + 1$ with $k_m \log m = o(m)$. Mark k_m rows $Y_1, \dots, Y_{k_m}$, independent of the remaining rows. Let $\mathcal{S}_m$ be the family of $(m-1)$-subsets of the unmarked rows, and write

$$L_m = |\mathcal{S}_m| = \binom{N_m - k_m}{m-1}.$$

For $S \in \mathcal{S}_m$, let u_S be a unit normal to the span of the rows indexed by S, whenever that span has dimension $m-1$. Put

$$a_S(t) = \mathbb{P}\left\{ \sum_{j=1}^{k_m} |\langle Y_j, u_S \rangle|^2 \le t^2 \,\middle|\, u_S \right\}.$$

For two subsets $S, T \in \mathcal{S}_m$, put

$$b_{S,T}(t) = \mathbb{P}\left\{ \sum_{j=1}^{k_m} |\langle Y_j, u_S \rangle|^2 \le t^2,\ \sum_{j=1}^{k_m} |\langle Y_j, u_T \rangle|^2 \le t^2 \,\middle|\, u_S, u_T \right\}.$$

Let S, T denote independent uniform elements of $\mathcal{S}_m$, and set

$$D(S,T) = m - |S \cap T|.$$

The following formulation is intentionally averaged and weighted. This avoids assuming that the random one-normal probabilities $a_S(t)$ are uniformly comparable for all S.

Theorem 8.1 (Second-moment universality criterion)**.** *Assume that N_m and r_m are integer-valued,*

$$\frac{N_m}{m} \to \gamma > 1, \qquad k_m = r_m + 1 \in \mathbb{N}, \qquad k_m \log m = o(m), \qquad 0 \le r_m \le N_m - m \quad \text{for all large } m.$$

Suppose that there are events $\mathcal{E}_m$, depending only on the unmarked rows, with $\mathbb{P}(\mathcal{E}_m) \to 1$, on which all spans appearing below have codimension one. For $t > 0$, define

$$\alpha_m(t) = \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_m} a_S(t)\right],$$

and, for a fixed constant $C_0 > 2$,

$$\beta_m^{\text{good}}(t) = \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_m} b_{S,T}(t) \mathbf{1}_{\{D(S,T) \ge C_0 k_m\}}\right],$$

$$\beta_m^{\text{bad}}(t) = \mathbb{E}\left[\mathbf{1}_{\mathcal{E}_m} b_{S,T}(t) \mathbf{1}_{\{2 \le D(S,T) < C_0 k_m\}}\right].$$

Assume that for every fixed $\varepsilon > 0$, at

$$t_m = \exp\left[-(h(\gamma) - \varepsilon)\frac{m}{k_m}\right],$$

the following estimates hold.

(U1) Averaged one-normal lower bound.

$$\alpha_m(t_m) \ge \exp\{-o(m)\} t_m^{k_m}.$$

More generally, the same lower bound is assumed uniformly throughout the window

$$\exp\{-O(m/k_m)\} \le t \le 1.$$

(U2) Weighted pair-normal second-moment bound.

$$\beta_m^{\mathrm{good}}(t_m) \le (1+o(1))\alpha_m(t_m)^2, \qquad \beta_m^{\mathrm{bad}}(t_m) = o(\alpha_m(t_m)^2).$$

Then

$$\limsup_{m\to\infty} \frac{k_m}{m} \log M_{m,r_m}^{\mathbb{R}}(A_m) \le -h(\gamma)$$

in probability.

If, in addition, the one-matrix lower-tail estimate

$$\mathbb{P}\{\sigma_m(B_m) \le t\} \le \exp\{o(m)\} t^{k_m}$$

holds uniformly for every $(m+r_m)\times m$ *row submatrix* B_m *and uniformly in the same exponential window, then*

$$\frac{k_m}{m} \log M_{m,r_m}^{\mathbb{R}}(A_m) \xrightarrow{P} -h(\gamma).$$

Proof. We first prove the upper bound. Define

$$X_S(t) = \mathbf{1}\left\{ \sum_{j=1}^{k_m} |\langle Y_j, u_S\rangle|^2 \le t^2 \right\}, \qquad Z_t = \sum_{S\in\mathcal{S}_m} X_S(t),$$

and truncate to the good event by

$$\widetilde{Z}_t = \mathbf{1}_{\mathcal{E}_m} Z_t.$$

The deterministic implication Lemma 4.1 uses only linear algebra, so $\widetilde{Z}_t > 0$ still implies

$$M_{m,r_m}^{\mathbb{R}}(A_m) \le t.$$

Let $t = t_m$. Since S is uniform in $\mathcal{S}_m$,

$$\mu_m := \mathbb{E}\widetilde{Z}_t = L_m \alpha_m(t).$$

By Lemma 2.1 and (U1),

$$\mu_m \ge \exp\{h(\gamma)m + o(m)\} \exp\{-o(m)\} t_m^{k_m} = \exp\{\varepsilon m - o(m)\} \to \infty.$$

We now bound the second moment. The diagonal contribution is

$$\sum_{S\in\mathcal{S}_m} \mathbb{E}[\mathbf{1}_{\mathcal{E}_m} X_S(t)] = L_m \alpha_m(t) = \mu_m.$$

For $S \ne T$, conditioning on the unmarked rows gives

$$\mathbb{E}[\mathbf{1}_{\mathcal{E}_m} X_S(t) X_T(t)] = \mathbb{E}[\mathbf{1}_{\mathcal{E}_m} b_{S,T}(t)].$$

Because $D(S,T) \ge 2$ for distinct S, T, the off-diagonal terms are split into the good and bad overlap regimes. Hence

$$\mathbb{E}\widetilde{Z}_t^2 \le \mu_m + L_m^2 \left(\beta_m^{\mathrm{good}}(t) + \beta_m^{\mathrm{bad}}(t) \right).$$

Dividing by $\mu_m^2 = L_m^2 \alpha_m(t)^2$ and using (U2) gives

$$\frac{\mathbb{E}\widetilde{Z}_t^2}{\mu_m^2} \le \frac{1}{\mu_m} + 1 + o(1) = 1 + o(1).$$

Thus $\mathrm{Var}(\widetilde{Z}_t) = o(\mu_m^2)$, and Chebyshev's inequality yields

$$\mathbb{P}\{\widetilde{Z}_t = 0\} \to 0.$$

Therefore

$$\mathbb{P}\{M^{\mathbb{R}}_{m,r_m}(A_m) > t_m\} \to 0,$$

which proves the asserted upper bound after letting $\varepsilon \downarrow 0$.

For the lower bound, fix $\varepsilon > 0$ and set

$$s_m = \exp\left[-(h(\gamma)+\varepsilon)\frac{m}{k_m}\right].$$

The union bound and the one-matrix estimate give

$$\begin{aligned}\mathbb{P}\{M^{\mathbb{R}}_{m,r_m}(A_m) \le s_m\} &\le \binom{N_m}{m+r_m}\exp\{o(m)\}s_m^{k_m}\\ &\le \exp\{h(\gamma)m + o(m) - (h(\gamma)+\varepsilon)m\} \to 0.\end{aligned}$$

This gives the matching lower bound and completes the proof. □

If the stronger pointwise comparability

$$a_S(t) = \exp\{o(m)\}\alpha_m(t)$$

is available uniformly in S, then the weighted assumptions above may be verified through the simpler ratio bounds involving

$$\frac{b_{S,T}(t)}{a_S(t)a_T(t)}.$$

The theorem itself, however, does not require such comparability.

8.2 Smooth iid entries

For iid entries with a bounded smooth density, the expected real theorem is

$$\frac{k_m}{m}\log M^{\mathbb{R}}_{m,r_m}(A_m) \xrightarrow{P} -h(\gamma),$$

with the corresponding complex limit $-h(\gamma)/2$. Several one-matrix results point in this direction. Tao and Vu proved universality of the least singular value distribution for iid matrices under finite-moment assumptions [17]. Rudelson and Vershynin proved sharp lower bounds for the smallest singular value of iid rectangular matrices [14]. These results address one matrix at a time. The present extremal problem asks for the minimum over exponentially many highly dependent row submatrices, which requires uniform control over random normals generated by all $(m-1)$-row spans.

A route to proving this universality is the following.

(a) Prove that all relevant normals u_S are delocalized, for example

$$\|u_S\|_\infty \le m^{-1/2+o(1)},$$

outside an exponentially negligible exceptional family of subsets S.

(b) Prove density-level local central limit estimates for $\langle X, u_S \rangle$ uniformly over such delocalized u_S at scales $t = e^{-\Theta(m/k_m)}$.

(c) Prove the corresponding two-normal density bound for pairs (u_S, u_T), with the correct determinant dependence on their angle.

(d) Insert these estimates into Theorem 8.1.

The phrase "density-level" is essential. Ordinary Berry–Esseen estimates produce distribution-function errors that are much larger than $e^{-\Theta(m)}$ in the fixed-r regime.

Conjecture 8.2 (Smooth iid universality)**.** *Let $A_m \in \mathbb{R}^{N_m \times m}$ have iid entries of mean zero and variance one, with a bounded sufficiently smooth density and sufficiently many finite moments. If $N_m/m \to \gamma > 1$, then for every fixed $r \geq 0$,*

$$\frac{1}{m} \log M^{\mathbb{R}}_{m,r}(A_m) \xrightarrow{P} -\frac{h(\gamma)}{r+1}.$$

For complex iid entries with a bounded two-real-dimensional density, the limit should be

$$-\frac{h(\gamma)}{2(r+1)}.$$

8.3 Bounded densities and atomic laws

Continuity prevents some exact singularities but does not by itself supply the upper-bound second moment. For square submatrices, the lower bound under bounded densities follows from known small-ball estimates for the least singular value. In contrast, atomic laws can have many exactly singular submatrices for arithmetic reasons. For Bernoulli sign matrices at $N = 2m - 1$, any fixed $m \times m$ submatrix is singular if two of its rows are equal. This event has probability at least 2^{-m} up to polynomial factors, and the number of square submatrices is of order $4^m/\sqrt{m}$. Thus exact singular submatrices can occur for reasons unrelated to the continuous hard edge.

A useful intermediate class is a smoothed discrete law,

$$a_{ij} = \xi_{ij} + \tau g_{ij}, \qquad \tau > 0,$$

where ξ_{ij} is discrete and g_{ij} is Gaussian. Such laws have continuous densities, so one-matrix lower-tail estimates are protected. The matching upper estimate should follow from Theorem 8.1 once the required uniform delocalization and density-level local estimates are established.

9 Further applications and consequences

9.1 Robust complement property

The complement property is qualitative: it asks whether every partition has a spanning side. The parameter $\omega(A)$ is quantitative: it asks how close one can come to violating the complement property. Theorem 1.7 shows that in random real phase retrieval near the critical threshold, the robust complement property has a hard-edge law. For $N = 2m - 1 + r$ with fixed r, the weakest spanning complement has least singular value

$$4^{-m/(r+1)+o_P(m)}.$$

The dangerous obstruction is an extreme hard-edge event among exponentially many almost-critical subframes: $m - 1$ rows create a hyperplane whose normal direction is simultaneously small on the remaining $r + 1$ marked rows.

9.2 Design rule for extra measurements

Suppose one wants the Balan–Wang conditioning scale $1/\omega(A_m)$ to be no worse than polynomial in m. Corollary 1.8 proves that this cannot happen in the Gaussian/radial critical-plus-redundancy model if

$$r_m + 1 = o(m/\log m).$$

Thus extra measurements must be at least of order $m/\log m$ before polynomial conditioning is even plausible. This is a necessary scale, not a sufficient theorem. At $r_m \asymp m/\log m$, the polynomial factors hidden in the hard-edge and overlap estimates become visible and should influence the exact polynomial exponent.

9.3 Conditioning of random polytopes and algorithms

Related conditioning estimates appear in the smoothed analysis of algorithms, for example in work of Rademacher and Shu on Frank–Wolfe methods [13]. The quantity $M_{m,r}$ is a worst-case conditioning parameter over row-selected nearly square subproblems. The theorem says that, in the Gaussian and radial settings covered here, the worst selected subproblem has a deterministic exponential law depending only on the row entropy $h(\gamma)$, the scalar dimension $d_{\mathbb{F}}$, and the surplus parameter $r+1$.

10 Open problems

Question 10.1 (Sharp transition around $m/\log m$)**.** *For real Gaussian phase retrieval with $N = 2m - 1 + r_m$, determine $\omega(A_m)$ when $r_m \asymp m/\log m$. Does $1/\omega(A_m)$ have a polynomial limit exponent depending on*

$$\lim \frac{r_m \log m}{m}?$$

Question 10.2 (Full smooth iid universality)**.** *Prove Conjecture 8.2. The central missing input is a uniform density-level local central limit theorem for random normals generated by all $(m-1)$-row spans.*

Question 10.3 (Atomic and conditioned discrete ensembles)**.** *For Bernoulli sign matrices, what is the correct analogue of $M_{m,r}$ after conditioning on full spark or after excluding exactly singular submatrices? Does a smoothed Bernoulli ensemble recover the continuous base immediately for every fixed smoothing level $\tau > 0$?*

Question 10.4 (Complex phase retrieval)**.** *The complex random matrix theorem gives the hard-edge factor $d_{\mathbb{C}} = 2$, but complex phase retrieval has a different injectivity threshold and more complicated algebraic geometry. Is there an analogue of Theorem 1.7 for complex phaseless measurements at the generic complex threshold?*

11 Conclusion

The square-submatrix theorem of [16] identifies the Gaussian critical base 1/4 in the Balan–Wang phase retrieval stability problem. The present extension shows that this base is part of a larger redundancy law. For Gaussian row submatrices with surplus r,

$$M^{\mathbb{F}}_{m,r}(G_m) = \exp\left\{-\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}m + o_P(m)\right\}.$$

The same law persists for rotationally invariant rows with subexponential radii and for subexponentially anisotropic Gaussian rows. In real phase retrieval with $N = 2m - 1 + r$,

$$\omega(A_m) = 4^{-m/(r+1)+o_P(m)}.$$

More generally, if $r_m + 1$ grows but $(r_m + 1)\log m = o(m)$, then

$$\frac{r_m + 1}{m} \log \omega(A_m) \xrightarrow{P} -\log 4.$$

Thus fixed redundancy improves the exponent exactly, while slowly growing redundancy still leaves super-polynomial instability. The proof also isolates the path toward smooth iid universality: one must replace Gaussian rotational invariance by uniform delocalization and density-level small-ball estimates for the random hyperplane normals.

A Auxiliary estimates used in the hard-edge and pair-normal proofs

This appendix records two elementary estimates used above. They are included to make the dependence on k explicit.

Lemma A.1 (A gamma-ratio bound). *Fix $b \in \{1/2, 1\}$. There is a constant C_b such that for all integers $m \geq 2$ and $1 \leq k \leq m$,*

$$m \frac{\Gamma(b(k+m))}{\Gamma(bk)\Gamma(b(k+1))\Gamma(1+bm)} \leq \exp\{C_b k \log m\}.$$

Proof. By Stirling's formula, there is a constant C_b such that for all $x \geq 1/2$,

$$C_b^{-1} x^{x-1/2} e^{-x} \leq \Gamma(x) \leq C_b x^{x-1/2} e^{-x}.$$

Applying this to the numerator and to $\Gamma(1 + bm)$ gives the crude bound

$$\frac{\Gamma(b(k+m))}{\Gamma(1+bm)} \leq (C_b m)^{bk}$$

for $1 \leq k \leq m$; when $bk < 1$ the left side is actually $O(1)$, so the same bound remains valid. The factor m is at most $e^{k \log m}$, and the denominator $\Gamma(bk)\Gamma(b(k+1))$ is bounded below by a positive constant depending only on b for all $k \geq 1$. This proves the claim after adjusting C_b. □

Lemma A.2 (Beta moment bound). *Let $R \sim \mathrm{Beta}(a, b)$ with $a \in \{1/2, 1\}$ and $b \geq 1$. If $0 \leq \alpha \leq b/2$, then*

$$\mathbb{E}(1-R)^{-\alpha} \leq \exp\left\{C_a \frac{\alpha}{b}\right\}.$$

Proof. The beta integral gives

$$\mathbb{E}(1-R)^{-\alpha} = \frac{\Gamma(b-\alpha)\Gamma(a+b)}{\Gamma(b)\Gamma(a+b-\alpha)}.$$

Taking logarithms and differentiating under the integral form of the difference,

$$\log \mathbb{E}(1-R)^{-\alpha} = \int_0^\alpha [\psi(a+b-s) - \psi(b-s)]\, ds.$$

For $0 \le s \le \alpha$, $b - s \ge b/2$. The digamma series identity

$$\psi(x+a) - \psi(x) = \int_0^\infty \frac{e^{-xt}(1-e^{-at})}{1-e^{-t}}\,dt$$

for $x > 0$ implies

$$0 \le \psi(x+a) - \psi(x) \le \frac{C_a}{x}.$$

Therefore the integrand is at most C_a/b, and the result follows. □

B Deterministic inequalities for phaseless stability

Let

$$\mathcal{A}_A(x) = |Ax|, \qquad x \in \mathbb{R}^m/\{\pm 1\},$$

and equip the quotient with the metric

$$d(x,y) = \min\{\|x-y\|_2\,, \|x+y\|_2\}.$$

Define the lower Lipschitz constant

$$\alpha(A) = \inf_{x \not\equiv \pm y} \frac{\||Ax| - |Ay|\|_2}{d(x,y)}.$$

Balan–Wang stability parameters are formulated in terms of singular values of complementary subframes. The following elementary inequality is the only deterministic input needed in this paper: the quantity $\omega(A)$ is an upper bound for the possible lower Lipschitz scale.

Lemma B.1. *For every real $N \times m$ matrix A,*

$$\alpha(A) \le \omega(A),$$

where

$$\omega(A) = \inf_{\substack{J \subset [N]:\, \operatorname{rank}(A_J) < m \\ |J^c| \ge m}} \sigma_m(A_{J^c}),$$

with the convention $\omega(A) = +\infty$ if the admissible family is empty.

Proof. If the admissible family is empty, there is nothing to prove. Fix an admissible $J \subset [N]$, so $\operatorname{rank}(A_J) < m$ and $|J^c| \ge m$. Choose a unit vector $u \in \ker A_J$, and choose a unit vector v satisfying

$$\|A_{J^c} v\|_2 = \sigma_m(A_{J^c}).$$

Set

$$x = \frac{u+v}{2}, \qquad y = \frac{v-u}{2}.$$

Then $x - y = u$, $x + y = v$, and hence $d(x,y) = 1$. For $i \in J$, we have $\langle a_i, u\rangle = 0$, so

$$|\langle a_i, x\rangle| = |\langle a_i, y\rangle|.$$

For $i \in J^c$, the real scalar identity

$$\big||p+q| - |q-p|\big| \le 2|q|$$

with $p = \langle a_i, u\rangle$ and $q = \langle a_i, v\rangle$ gives

$$\big||\langle a_i, x\rangle| - |\langle a_i, y\rangle|\big| \le |\langle a_i, v\rangle|.$$

Therefore

$$\||Ax| - |Ay|\|_2 \le \|A_{J^c} v\|_2 = \sigma_m(A_{J^c}).$$

Taking the infimum over admissible J gives $\alpha(A) \le \omega(A)$. $\square$

For a full-spark matrix with $N = 2m - 1 + r$, the largest rank-deficient side has $m - 1$ rows, and its complement has $m + r$ rows, so the admissibility condition $|J^c| \ge m$ is automatic for every rank-deficient side. Lemma 7.2 therefore identifies $\omega(A)$ with the row-submatrix hard-edge quantity $M^{\mathbb{R}}_{m,r}(A)$. At $r = 0$ this is exactly the barely spanning complement obstruction at the real injectivity threshold. For $r > 0$, it is still a deterministic obstruction to the lower Lipschitz constant, but the paper does not assert that it equals the full Balan–Wang partition minimum over all spanning–spanning partitions.

C Examples covered by the comparison theorems

Uniform spherical measurements. If every row is sampled uniformly from the real or complex unit sphere, then $R_i \equiv 1$ and the radial condition is automatic. Hence the Gaussian fixed- and moderate-surplus exponents hold for spherical measurements.

Random radii. Suppose $a_i = R_i \Theta_i$ and, for every fixed $\eta > 0$,

$$\mathbb{P}\{R_i \ge e^{\eta m/k_m}\} + \mathbb{P}\{R_i \le e^{-\eta m/k_m}\} = o(m^{-1})$$

uniformly in i. A union bound gives the radial condition. In particular, any fixed distribution supported in $[c, C]$ with $0 < c < C < \infty$ is allowed, as are lognormal radii whose logarithmic tails are negligible at the scale m/k_m.

Elliptical covariance. The elliptical theorem allows deterministic covariance matrices satisfying

$$\max\{|\log \lambda_{\min}(\Sigma_m)|,\, |\log \lambda_{\max}(\Sigma_m)|\} = o(m/k_m).$$

For fixed r, this permits any polynomial condition number and many subexponential condition numbers. If a covariance direction has exponentially small variance at the hard-edge scale, the exponent can shift by the covariance scale.

D A proof template for smooth iid universality

We spell out the smooth iid strategy in a form compatible with Theorem 8.1. Let the entries be iid with a bounded smooth density f, mean zero, variance one, and sufficiently many moments.

Step 1: no-gaps delocalization of normals. For each $(m-1)$-row set S, let u_S be a unit normal. One needs a high-probability event on which, uniformly for all but an exponentially negligible family of S,

$$\|u_S\|_\infty \le m^{-1/2+\eta_m}, \qquad \eta_m \to 0.$$

This is stronger than delocalization for one random null vector because it must hold over exponentially many row subsets.

Step 2: density local central limit theorem. For a deterministic delocalized vector u, the linear form

$$\langle X, u\rangle = \sum_{j=1}^{m} X_j u_j$$

should have a density f_u satisfying

$$f_u(0) = \frac{1}{\sqrt{2\pi}} e^{o(m)}, \qquad \sup_x f_u(x) \le e^{o(m)}.$$

At the fixed-surplus scale, one needs density control down to intervals of length $e^{-\Theta(m)}$, not merely distribution-function approximation.

Step 3: two-normal density bound. For two delocalized vectors u, v, the pair

$$(\langle X, u\rangle, \langle X, v\rangle)$$

should have a joint density near the origin bounded by

$$e^{o(m)}(1 - \langle u, v\rangle^2)^{-1/2}$$

in the real case, with the corresponding determinant factor in the complex case. This reproduces the Gaussian pair-slab calculation up to subexponential factors.

Step 4: insert into the second moment. With these one- and two-normal estimates, the proof of Proposition 4.5 runs with $e^{o(m)}$ losses. The overlap combinatorics is distribution-free. The lower bound follows from a rectangular least-singular-value tail for iid matrices with bounded density. This is precisely the structure of Theorem 8.1.

Funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon 2022 research and innovation programme (grant agreement No. 101041711), by the Simons Foundation, by Heights Labs, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25).

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the author used generative AI tools for language editing, proof organization, and bibliographic formatting assistance. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the mathematical content and final manuscript.

Declaration of competing interest

The author declares no competing interests.

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